Supplementary Material

Multi-Scale Workload Similarity Measurement for Proactive Database Index Tuning Using Hierarchical Signal Processing

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This document provides complete proofs for the four supporting lemmas (L1–L4) and the six core theorems (T1–T6) stated in Section IV of the main paper, followed by extended experimental details (Section XI) including full A1–A7 results, A-CE protocol, extended proof sketches, remarks, and deployment guidelines. Notation follows the main paper exactly. The five HSM dimensions are: S_R (query-template rank correlation via Spearman's ρ_s, mapped through $1-\arccos(\rho_s)/\pi$), S_V (volume ratio), S_T (query-template angular distance, $1-(2/\pi)\arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B)$), S_A (attribute-set Jaccard), S_P (DWT-based temporal pattern). All five similarity functions lie in $[0,1]$; the induced distances $d_i = 1 - S_i$ all satisfy the triangle inequality analytically (Lemma 1, property P4). The composite score is $S_{\text{HSM}}(W_A, W_B) = \sum_i w_i S_i(W_A, W_B)$ with default weights $w_R = 0.25$, $w_V = w_T = w_A = 0.20$, $w_P = 0.15$. $d_{\text{HSM}} = 1 - S_{\text{HSM}}$ denotes the corresponding distance. $N_{\text{pts}} = |W_A| + |W_B|$ denotes the total query events across both windows.

I. Proof of Lemma 1 (HSM Metric Properties)

Lemma 1 (HSM Metric Properties). *$\forall W_i, W_j, W_k \in \mathcal{W}$:*

$$
\begin{aligned}
&(P1)\ S_{\text{HSM}}(W_i, W_j) \in [0,1];\\
&(P2)\ S_{\text{HSM}}(W_i, W_i) = 1;\\
&(P3)\ S_{\text{HSM}}(W_i, W_j) = S_{\text{HSM}}(W_j, W_i);\\
&(P4)\ 1-S_{\text{HSM}}(W_i, W_k) \le (1-S_{\text{HSM}}(W_i, W_j))\\
&\qquad + (1-S_{\text{HSM}}(W_j, W_k)).
\end{aligned}
$$

Proof. We verify each property in turn.

P1 (Non-negativity and boundedness). By construction, each component score $S_d(W_A, W_B) \in [0,1]$. $S_R = 1 - \arccos(\rho_s(\mathbf{r}_A, \mathbf{r}_B))/\pi \in [0,1]$ since $\rho_s \in [-1,1]$ implies $\arccos(\rho_s) \in [0,\pi]$. $S_V = \min(|W_A|, |W_B|)/\max(|W_A|, |W_B|) \in [0,1]$. $S_T = 1 - (2/\pi)\arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B) \in [0,1]$: since query-type count vectors are non-negative, $\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B \in [0,1]$, so $\arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B) \in [0,\pi/2]$, giving $(2/\pi)\arccos(\cdot) \in [0,1]$ and $S_T \in [0,1]$. $S_A = |A_A \cap A_B|/|A_A \cup A_B| \in [0,1]$ (Jaccard index). $S_P \in [0,1]$ by normalised FastDTW construction. Therefore $S_{\text{HSM}} = \sum w_i S_i \in [0,1]$ since $w_i > 0$, $\sum w_i = 1$, each $S_i \in [0,1]$.

P2 (Identity of indiscernibles). For $W_A = W_B = W$: $S_R(W,W) = 1 - \arccos(1)/\pi = 1$ since $\rho_s = 1$ for identical rank vectors; $S_V(W,W) = |W|/|W| = 1$; $S_T(W,W) = 1 - (2/\pi)\arccos(1) = 1$; $S_A(W,W) = |A \cap A|/|A \cup A| = 1$; $S_P(W,W) = 1$ since FastDTW distance between identical sequences is 0. Therefore $S_{\text{HSM}}(W,W) = \sum w_i \times 1 = 1$.

P3 (Symmetry). Each component is symmetric by construction: $\rho_s(\mathbf{r}_A, \mathbf{r}_B) = \rho_s(\mathbf{r}_B, \mathbf{r}_A)$; min/max are order-independent; $\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B = \hat{\mathbf{v}}_B \cdot \hat{\mathbf{v}}_A$; Jaccard: $|A_A \cap A_B| = |A_B \cap A_A|$; FastDTW satisfies $\text{DTW}(X,Y) = \text{DTW}(Y,X)$. Hence $S_{\text{HSM}}(W_A, W_B) = S_{\text{HSM}}(W_B, W_A)$.

P4 (Triangle inequality for d_{HSM}). Write $d_{\text{HSM}} = 1 - S_{\text{HSM}} = \sum w_i(1 - S_i) = \sum w_i d_i$ where $d_i = 1 - S_i$. We prove TI analytically for all five component distances.

$d_V = 1 - S_V$. A ratio distance on positive reals; TI holds by the standard ratio-metric argument (Deza & Deza, 2009, §4.1).

$d_A = 1 - S_A$. The Jaccard distance on finite sets, a well-known metric [3].

$d_P = 1 - S_P$. DTW on normalised sequences satisfies TI; P4 holds within the approximation error ε bounded in Theorem 1 (Section V).

$d_T = (2/\pi)\arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B)$. The angular (geodesic) distance on $\mathbb{S}^{n-1}$, satisfying TI by the triangle inequality for geodesics on a sphere [3].

$d_R = \arccos(\rho_s)/\pi$. Spearman's ρ_s equals the Pearson correlation of rank vectors. Pearson correlation $r \in [-1,1]$ is related to the correlation angle $\phi = \arccos(r)$, which satisfies TI on the unit sphere (Kendall & Gibbons, 1990, §1.9). Therefore $d_R = \phi/\pi$ also satisfies TI (positive scaling preserves TI).

Composite. Since each d_i satisfies TI and $w_i > 0$, the weighted sum $d_{\text{HSM}} = \sum w_i d_i$ satisfies:

$$
\begin{aligned}
d_{\text{HSM}}(W_i, W_k) &= \sum_d w_d\, d_d(W_i, W_k)\\
&\le \sum_d w_d[d_d(W_i, W_j) + d_d(W_j, W_k)]\\
&= d_{\text{HSM}}(W_i, W_j) + d_{\text{HSM}}(W_j, W_k).
\end{aligned}
$$

This establishes P4. □ □

II. Proof of Lemma 2 (DWT Parameter Selection)

Lemma 2 (DWT Parameter Selection). *Let k be the number of canonical workload states (workload window model, Section III of the main paper). The minimum SAX alphabet size α^* guaranteeing correct state discrimination under the maximum-entropy model and the two-sided burst extension satisfies $\alpha^* = 2k$. With $k = 4$: $\alpha^* = 8$. Under Quantile-SAX breakpoints, $P(\text{misclass.}) < 1/\alpha = 1/8$.*

Proof. **Step 1 (State-space cardinality).** The workload window model defines $k = 2 \times 2 = 4$ canonical states by crossing two binary attributes: volume regime ($\langle\text{high}\rangle$ vs. $\langle\text{low}\rangle$) and query-type dominance ($\langle\text{read}\rangle$ vs. $\langle\text{write}\rangle$). The two-sided burst extension augments each state with a burst direction (onset vs. recovery), giving $2k = 8$ discriminable regions on the standardised QPS axis.

Step 2 (Minimum alphabet). An alphabet of size α creates α bins on the real line. Correct discrimination requires each of the 8 regions to map to a unique bin, so $\alpha \ge 2k = 8$ is necessary. Sufficiency: set $\alpha = 8$. Under Quantile SAX with $N(0,1)$ breakpoints, the eight bins partition the z-score axis into eight equiprobable regions of probability $1/\alpha$ each. Each burst region maps one-to-one to exactly one bin.

Step 3 (Misclassification bound). Under the maximum-entropy (worst-case) distribution, the probability mass within any single bin is exactly $1/\alpha$. A query window is misclassified only if its z-score falls in an adjacent bin, which has probability at most $1/\alpha$. Hence $P(\text{misclassification}) \le 1/\alpha = 1/8$.

Remark 1. *The implementation uses $\alpha = 4$, a computationally efficient practical choice. $\alpha^* = 8$ provides the theoretical worst-case guarantee. Section V of the main paper reports empirical accuracy under $\alpha = 4$.*

□ □

III. Proof of Lemma 3 (Tight Linear Complexity)

Lemma 3 (Tight Linear Complexity)**.** *$T_{\mathrm{HSM}}(N_{\mathrm{pts}}, N) = \Theta(N_{\mathrm{pts}})$ in both upper and lower bound, independent of database cardinality N. Space complexity is $\Theta(N_{\mathrm{pts}})$. Moreover, the lower bound holds for any function $\mathrm{sim} : \mathcal{W} \times \mathcal{W} \to [0, 1]$ obeying P1–P4: such a function must inspect $\Omega(N_{\mathrm{pts}})$ elements of each input window.*

Proof. We prove the upper and lower bounds in turn.

Upper bound: $T_{\mathrm{HSM}} = O(N_{\mathrm{pts}})$. We bound the cost of each dimension.

S_R: a single pass over W_A and W_B to build query-frequency rank vectors $\mathbf{r}_A, \mathbf{r}_B$ (size $k \ll N_{\mathrm{pts}}$, fixed template set); rank correlation costs $O(k \log k) = O(1)$. Total $\Theta(N_{\mathrm{pts}})$.

S_V: compare $|W_A|$ and $|W_B|$ (window-size counters); cost $O(1)$.

S_T: accumulate the k-element template-frequency vector during the S_R pass; normalise; compute $\arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B)$. With $k \ll N_{\mathrm{pts}}$ fixed, additional cost is $O(k) = O(1)$ per window, $O(N_{\mathrm{pts}})$ total.

S_A: collect attribute sets A_A, A_B ($\leq N_{\mathrm{pts}}$ total attributes); hash-based Jaccard yields $O(|A_A| + |A_B|) = O(N_{\mathrm{pts}})$.

S_P: db4 DWT at level $L = 3$ in $O(N_{\mathrm{pts}})$ [2]; SAX encoding in $O(N_{\mathrm{pts}})$; FastDTW with Sakoe–Chiba radius r on sub-band vectors of length N_{pts} in $O(N_{\mathrm{pts}})$ [9].

Total: $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}}) + O(1) + O(N_{\mathrm{pts}}) + O(N_{\mathrm{pts}}) + O(N_{\mathrm{pts}}) = \Theta(N_{\mathrm{pts}})$. No operation accesses database pages, so the cost is independent of N. Each dimension stores $O(N_{\mathrm{pts}})$ intermediate values, giving space $\Theta(N_{\mathrm{pts}})$.

Lower bound: $T_{\mathcal{A}} = \Omega(N_{\mathrm{pts}})$ for any P1–P4 similarity. Suppose for contradiction that some algorithm $\mathcal{A}$ computing $\mathrm{sim} : \mathcal{W} \times \mathcal{W} \to [0, 1]$ satisfies P1–P4 yet reads only $k < N_{\mathrm{pts}}$ elements of each window.

Construct an adversary α producing two window pairs (W, W') and (W, W''). Both W' and W'' agree with W on all k elements that $\mathcal{A}$ reads, but differ on one unread query event q^* at some unread position:

- W' contains q^* with attribute set A';
- W'' contains q^* with attribute set $A'' \neq A'$ chosen so that $A_W \cup A' \neq A_W \cup A''$.

Since $\mathcal{A}$ does not read q^*, it must return the same score on both pairs: $\mathcal{A}(W, W') = \mathcal{A}(W, W'')$.

However, $S_A(W, W') \neq S_A(W, W'')$ because the Jaccard overlap on attribute sets differs. Any P1–P4 similarity that is sensitive to the attribute dimension must therefore distinguish these two pairs, and $\mathcal{A}$ is incorrect on at least one of them. This contradicts the assumption. Hence $T_{\mathcal{A}} = \Omega(N_{\mathrm{pts}})$. Combined with the upper bound, $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ is asymptotically optimal. □ □

IV. Proof of Lemma 4 (Wavelet Coefficient Bound)

Lemma 4 (Wavelet Coefficient Bound)**.** *For any signal $q(t)$ defined on $[0, N_{\mathrm{pts}})$ with $\sup |q(t)| \leq M$, and its level-L db4 DWT approximation $\tilde{q}$,*

$$\|q - \tilde{q}\|_2 \;\leq\; C^L_{\mathrm{db4}}\, M \sqrt{N_{\mathrm{pts}}}, \qquad C_{\mathrm{db4}} = \frac{\sqrt{2}}{2} \approx 0.707.$$

Proof. For an orthonormal wavelet with low-pass filter constant C (defined via $\|h\|_2 = 1/\sqrt{2}$, i.e. $C = \sqrt{2}/2$ for db4), the level-L DWT approximation $\tilde{q}$ satisfies the Mallat truncation bound ([7], Theorem 7.13):

$$\|q - \tilde{q}\|_2 \;\leq\; C^L_{\mathrm{db4}}\, \|q\|_2.$$

Since $\sup |q(t)| \leq M$ and the discrete-time series has N_{pts} samples, $\|q\|_2 \leq M\sqrt{N_{\mathrm{pts}}}$. Substituting yields $\|q - \tilde{q}\|_2 \leq C^L_{\mathrm{db4}} M \sqrt{N_{\mathrm{pts}}}$. □ □

V. Proof of Theorem 1 (Wavelet Pattern Preservation)

Theorem 1 (Wavelet Pattern Preservation)**.** *Let $q(t)$ satisfy $\sup |q(t)| \leq M$. The level-L db4 DWT used by S_P introduces pattern-similarity error bounded by*

$$\varepsilon(M, N_{\mathrm{pts}}, L) \;=\; \frac{2\, C^L_{\mathrm{db4}}\, M}{\sqrt{N_{\mathrm{pts}}}}, \qquad C_{\mathrm{db4}} = \frac{\sqrt{2}}{2},$$

which is strictly decreasing in both L and N_{pts}.

Proof. **Step 1 (Wavelet truncation in ℓ^2).** By Lemma 4, $\|q - \tilde{q}\|_2 \leq C^L_{\mathrm{db4}} M \sqrt{N_{\mathrm{pts}}}$.

Step 2 (Impact on S_P). S_P is computed via FastDTW on SAX-encoded DWT sub-band vectors. The pattern-similarity score changes by at most

$$\begin{aligned} \left| S_P^{\mathrm{exact}} - S_P^{\mathrm{approx}} \right| &\leq \frac{2\, \|q - \tilde{q}\|_2}{N_{\mathrm{pts}}} \\ &\leq \frac{2\, C^L_{\mathrm{db4}}\, M}{\sqrt{N_{\mathrm{pts}}}} = \varepsilon(M, N_{\mathrm{pts}}, L). \end{aligned}$$

The factor 2 arises from comparing two windows, each contributing at most $\varepsilon/2$ to the normalised DTW distance.

Step 3 (Monotonicity). $C_{\mathrm{db4}} = \sqrt{2}/2 < 1$, so C^L_{db4} is strictly decreasing in L; $\varepsilon \propto 1/\sqrt{N_{\mathrm{pts}}}$ is strictly decreasing in N_{pts}. Both monotonicity claims follow directly from the closed form. □ □

Remark 2. *The operator can plug measured (M, N_{pts}, L) and the filter constant $C_{\mathrm{db4}} = \sqrt{2}/2$ directly into the closed form to certify approximation quality before deployment.*

VI. Proof of Theorem 2 (Minimal Sufficient Dimensions)

Theorem 2 (Minimal Sufficient Dimensions)**.** *$\forall\, d \in \{S_R, S_V, S_T, S_A, S_P\}$, $\exists\, (W_A, W_B), (W_C, W_D) \in \mathcal{W}$ such that: (i) $S_d(W_A, W_B) = S_d(W_C, W_D)$ (d is uninformative on this pair); (ii) $|S_{\mathrm{HSM}}(W_A, W_B) - S_{\mathrm{HSM}}(W_C, W_D)| \geq w_d$. HSM_{-d} is therefore a strictly weaker discriminator than HSM, so the five-dimensional decomposition is minimally sufficient.*

Proof. Let HSM_{-d} denote HSM with w_d redistributed equally ($w_i \leftarrow w_i/(1 - w_d)$ for $i \neq d$).

General construction. Fix the other-dimension scores to be identical for both pairs: $S_i(W_A, W_B) = S_i(W_C, W_D)$ for all $i \neq d$. Let $S_d(W_A, W_B) = 1$, $S_d(W_C, W_D) = 0$. Then:

$$S_{\mathrm{HSM}}(W_A, W_B) - S_{\mathrm{HSM}}(W_C, W_D) = w_d(1 - 0) = w_d.$$

$\mathrm{HSM}_{-d}(W_A, W_B) = \mathrm{HSM}_{-d}(W_C, W_D)$ since all remaining dimensions are equal, so HSM_{-d} cannot discriminate the two pairs while HSM separates them by w_d.

Explicit witness pairs (default weights $w_R = 0.25$; $w_V = w_T = w_A = 0.20$; $w_P = 0.15$):

S_R: (W_A, W_B) same query-template rank distribution ($\rho_s = 1$, $S_R = 1$); (W_C, W_D) perfectly anti-correlated ($\rho_s = -1$, $S_R = 0$). $|S_{\mathrm{HSM}}$ diff$| = w_R = 0.25$. ✓

S_V: (W_A, W_B) 100-query vs. 100-query ($S_V = 1$); (W_C, W_D) 100-query vs. 0-query ($S_V = 0$). $|S_{\mathrm{HSM}}$ diff$| = w_V = 0.20$. ✓

S_T: (W_A, W_B) identical template distributions ($\hat{\mathbf{v}}_A = \hat{\mathbf{v}}_B$, $S_T = 1$); (W_C, W_D) disjoint template sets (e.g. `Q1,Q3,Q6` vs. `Q10,Q11,Q18`, $S_T = 0$). $|S_{\mathrm{HSM}}$ diff$| = w_T = 0.20$. ✓

S_A: (W_A, W_B) identical attribute set $\{x, y\}$ ($S_A = 1$); (W_C, W_D) $\{x, y\}$ vs. $\{z, w\}$ ($S_A = 0$). $|S_{\mathrm{HSM}}$ diff$| = w_A = 0.20$. ✓

S_P: (W_A, W_B) identical temporal patterns ($S_P = 1$); (W_C, W_D) steady QPS vs. bursty QPS (FastDTW dist $\to$ max, $S_P = 0$). $|S_{\mathrm{HSM}}$ diff$| = w_P = 0.15$. ✓

In each case, HSM discriminates the witness pairs while HSM_{-d} cannot, confirming that removing any single dimension strictly reduces discriminative power.

Sufficiency. The five dimensions $(S_R, S_V, S_T, S_A, S_P)$ collectively span rank, magnitude, angular, attribute-set, and temporal information; no two are linearly equivalent on $\mathcal{W}$. Combined with necessity above, the decomposition is minimally sufficient: no four-dimensional subset is sufficient. □ □

VII. Proof of Theorem 3 (Economic Optimality of HSM Gating)

Theorem 3 (Economic Optimality of HSM Gating)**.** *Under the cost model of Table 2 in the main paper (system-measurable constants a, b, f, g, c), define $Q_{\min}(N) = (aN \log N + b)/(fN - g \log N)$. Then: (i) $Q_{\min}(N) = \Theta(\log N)$ as $N \to \infty$, and $C_{\mathrm{comp}}/C_{\mathrm{create}} \to 0$; (ii) for $Q > Q_{\min}(N)$, the optimal HSM decision threshold is $\theta^*(N, Q) = 1 - Q_{\min}(N)/Q$; (iii) HSM's per-window computation cost is $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ regardless of the access attribute $A(W)$; (iv) for any access-attribute-dependent method M (Definition 2 in the main paper), an attribute transition forces $\Theta(N \log N)$ reconstruction cost on M but only $O(|A|) = O(1)$ marker update cost on HSM, so over τ transitions HSM saves $\Omega(\tau \cdot N \log N)$; (v) $T_{\mathrm{HSM}}/T_M \to 0$ as $N \to \infty$.*

Proof. **Part (i): $Q_{\min}(N) = \Theta(\log N)$.** REBUILD yields positive net benefit when the gain from Q future queries with a fresh index exceeds the rebuild cost:

$$Q\,(c_{\mathrm{scan}} - c_{\mathrm{idx}}) \;\geq\; C_{\mathrm{create}} + C_{\mathrm{drop}}.$$

Substituting $c_{\mathrm{scan}} - c_{\mathrm{idx}} = fN - g \log N$ (dominant term fN) and $C_{\mathrm{create}} = aN \log N$, $C_{\mathrm{drop}} = b$:

$$Q_{\min}(N) = \frac{aN \log N + b}{fN - g \log N}.$$

Dividing numerator and denominator by N and taking $N \to \infty$: $Q_{\min}(N) \to (a \log N)/f = \Theta(\log N)$. Furthermore $C_{\mathrm{comp}}/C_{\mathrm{create}} = cN_{\mathrm{pts}}/(aN \log N) \to 0$ since N_{pts} is bounded (Lemma 3) while $N \log N \to \infty$.

Part (ii): Optimal threshold. The HSM score $h = S_{\mathrm{HSM}}(W_c, W_{\mathrm{ref}})$ measures how similar the current workload is to the reference workload at last build. The expected net benefit of RETAIN over REBUILD at score h is

$$\mathrm{NetBenefit}(h) = Q\,(c_{\mathrm{scan}} - c_{\mathrm{idx}})\,h - C_{\mathrm{comp}} - (1-h)\,C_{\mathrm{create}}.$$

Setting $\mathrm{NetBenefit}(\theta^*) = 0$ and using $C_{\mathrm{comp}} \ll C_{\mathrm{create}}$ from Part (i): $\theta^*(N, Q) = 1 - Q_{\min}(N)/Q$ for $Q > Q_{\min}(N)$. REBUILD is triggered when $h < \theta^*$, i.e. when the workload has drifted sufficiently that the index no longer fits economically.

Part (iii): HSM cost is access-attribute-independent. By Lemma 3, $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ with no database page access. The access-attribute set $A(W) = (\mathcal{T}(W), \mathcal{C}(W))$ is extracted as a by-product of the S_A computation, already counted in the $\Theta(N_{\mathrm{pts}})$ bound. The cost does not depend on which attributes are accessed.

Part (iv): Per-transition savings. When $A(W_c) \neq A(W_{c-1})$, HSM updates its attribute-set representation by a set-difference operation, costing $|A(W_c)\,\Delta\,A(W_{c-1})| \leq |A|$. For a fixed schema with $|A| = O(1)$, this is $O(1)$ additional work. By Definition 2 in the main paper, M is a physical index structure: when the access attribute set changes, the index must be rebuilt on the new column set, costing $\Theta(N \log N)$ [8]. Over τ attribute transitions: $\mathrm{Cost}_M = \tau \cdot \Theta(N \log N)$, while $\mathrm{Cost}_{\mathrm{HSM}} = \tau \cdot O(N_{\mathrm{pts}} + |A|) + \tau_{\mathrm{rebuild}} \cdot \Theta(N \log N)$ with $\tau_{\mathrm{rebuild}} \leq \tau$ the number of HSM-confirmed REBUILDs ($h < \theta^*$). Each suppressed false-alarm transition saves $\Theta(N \log N)$; with at least one such per run, savings amount to $\Omega(\tau \cdot N \log N)$.

Part (v): Asymptotic ratio.

$$\frac{T_{\mathrm{HSM}}}{T_M} = \frac{O(N_{\mathrm{pts}})}{\Theta(N \log N)} = O\left(\frac{N_{\mathrm{pts}}}{N \log N}\right) \to 0 \quad (N \to \infty),$$

since N_{pts} is the fixed window size and does not grow with N. □ □

Corollary (to Theorem 3, Access-Attribute Independence). Among all methods surveyed in Table 5 (main paper), HSM is the unique access-attribute-independent method. For any workload with at least one suppressed false-alarm transition, HSM achieves $\Theta(N \log N)$ net savings over any access-attribute-dependent baseline.

VIII. Proof of Theorem 4 (Detector-Quality Concentration via Stratified Hoeffding)

We reformulate HSM as a binary classifier of phase boundaries. For a threshold $\theta \in [0, 1]$, HSM decides RETAIN on a window pair iff $S_{\mathrm{HSM}} \geq \theta$. Two intrinsic quantities characterise the detector:

$$\mathrm{TPR}(\theta) = \Pr[S_{\mathrm{HSM}} \geq \theta \mid \text{within-phase}],$$
$$\mathrm{TNR}(\theta) = \Pr[S_{\mathrm{HSM}} < \theta \mid \text{cross-phase}].$$

Theorem 4 (Detector-Quality Concentration via Stratified Hoeffding)**.** *Let $\hat{\mathrm{TPR}}_{m_w} = (1/m_w) \sum_{i=1}^{m_w} \mathbf{1}[S_{\mathrm{HSM}}(W_i, W_i') \geq \theta]$ be the empirical TPR computed from m_w i.i.d. within-phase pairs, and let $\hat{\mathrm{TNR}}_{m_c} = (1/m_c) \sum_{j=1}^{m_c} \mathbf{1}[S_{\mathrm{HSM}}(W_j, W_j') < \theta]$ be the empirical TNR computed from m_c i.i.d. cross-phase pairs. Each indicator is bounded in $[0, 1]$. Then for any $\delta \in (0, 1)$, with $\eta(m, \delta) = \sqrt{\log(4/\delta)/(2m)}$,*

$$\Pr\Big[\,|\hat{\mathrm{TPR}}_{m_w} - \mathrm{TPR}| \leq \eta(m_w, \delta) \wedge |\hat{\mathrm{TNR}}_{m_c} - \mathrm{TNR}| \leq \eta(m_c, \delta)\,\Big] \geq 1 - \delta. \tag{1}$$

Proof. Let $X_i = \mathbf{1}[S_{\mathrm{HSM}}(W_i, W_i') \geq \theta]$ for $i = 1, \ldots, m_w$ be i.i.d. Bernoulli(TPR) random variables bounded in $[0, 1]$. By Hoeffding's inequality [4] applied at confidence $1 - \delta/2$,

$$\begin{aligned}&\Pr\big[\,|\hat{\mathrm{TPR}}_{m_w} - \mathrm{TPR}| > \eta(m_w, \delta)\,\big] \\ &\quad \leq 2\exp\big(-2m_w\,\eta(m_w, \delta)^2\big) = 2\exp\big(-\log(4/\delta)\big) = \delta/2.\end{aligned}$$

The same bound applies independently to $Y_j = \mathbf{1}[S_{\mathrm{HSM}}(W_j, W_j') < \theta]$ for $j = 1, \ldots, m_c$:

$$\Pr\big[\,|\hat{\mathrm{TNR}}_{m_c} - \mathrm{TNR}| > \eta(m_c, \delta)\,\big] \;\leq\; \delta/2.$$

By a union bound, the probability that *either* band fails is at most δ, so the joint event in the theorem statement holds with probability at least $1 - \delta$. □ □

Remark 3 (Why a stratified design is preferred over the mixture form)**.** *The legacy HSM analysis treated the window-level* RETAIN *indicator as a single Bernoulli draw from the mixture* $p_{retain} = \pi_w \mathrm{TPR} + (1 - \pi_w)(1 - \mathrm{TNR})$ *and applied a single Hoeffding bound. We make two distinct points in favour of the stratified design used in Theorem 4.*

(i) Variance reduction. *The Neyman-allocated stratified estimator with the same total budget* $m = m_w + m_c$ *has variance* $\pi_w^2 \mathrm{TPR}(1-\mathrm{TPR})/m_w + (1-\pi_w)^2 \mathrm{TNR}(1-\mathrm{TNR})/m_c$*, strictly less than the mixture variance* $p_{retain}(1-p_{retain})/m$ *whenever* $\mathrm{TPR} \neq 1 - \mathrm{TNR}$*. This variance gap propagates through Bernstein- and Bennett-type concentration bounds (which depend on variance), and through any asymptotic CLT-based confidence interval.*

(ii) Operational identifiability. *Hoeffding's bound itself uses only the range* $[0, 1]$*, not the variance, so the per-stratum Hoeffding bands of Theorem 4 are not strictly tighter than the mixture Hoeffding bound at every* m*. The decisive advantage of stratification is* operational*: it returns the two physically distinct quantities* $(\mathrm{TPR}, \mathrm{TNR})$ *that the operator actually needs in order to apply Theorem 5 at any deployment-specific phase-mix prior* π_w*, instead of a single mixture rate that conflates detector quality with workload prior. Without this decomposition, the same* p_{retain} *can arise from very different* $(\mathrm{TPR}, \mathrm{TNR})$ *pairs and the speedup prediction becomes ambiguous.*

Remark 4 (Worked example at SF = 3.0 (worst-case))**.** *At SF = 3.0,* $\theta = 0.75$ *on the scale traces of Section V, the empirical rates over* $m_w = m_c = 160$ *pairs are* $\hat{\mathrm{TPR}} = 0.9857$*,* $\hat{\mathrm{TNR}} = 0.9701$*. With* $\delta = 0.05$*,* $\eta = \sqrt{\log(80)/320} \approx 0.117$*, giving* $\mathrm{TPR} \in [0.869, 1.000]$ *and* $\mathrm{TNR} \in [0.853, 1.000]$ *at 95 % confidence. Increasing the per-stratum sample size to* $m_w = m_c = 1120$ *(the full set of pairs available in the existing 10-seed traces) shrinks* η *to* ≈ 0.044*, giving* $\mathrm{TPR} \in [0.941, 1.000]$ *and* $\mathrm{TNR} \in [0.926, 1.000]$.

IX. Proof of Theorem 5 (Operational Speedup with Phase-Mix Prior)

The original asymptotic gating result of HSM treated p_{stable} as a single opaque scalar. Once HSM is reformulated as a binary classifier (Theorem 4), p_{stable} decomposes into two independent factors: an *intrinsic* detector quality $(\mathrm{TPR}, \mathrm{TNR})$ and a *deployment-dependent* phase-mix prior $\pi_w = \Pr[\text{a window is within-phase}]$. Theorem 5 transports the Theorem 4 confidence band into a parameterised speedup curve.

Theorem 5 (Operational Speedup with Phase-Mix Prior)**.** *Let* $\mathcal{A}$ *be any index-management system with* $T_{\mathcal{A}}(N) = \Omega(N \log N)$*, and let*

$$p_{\mathrm{stable}}(\pi_w, \mathrm{TPR}, \mathrm{TNR}) = \pi_w \mathrm{TPR} + (1 - \pi_w)(1 - \mathrm{TNR}).$$

Define $\mathrm{Speedup}(N) = \mathrm{Cost}_{\mathrm{naive}}(N)/\mathrm{Cost}_{\mathrm{gated}}(N)$*. Then*

(i) $\mathrm{Speedup}(N) \to 1/(1 - p_{\mathrm{stable}})$ *and* $T_{\mathrm{HSM}}/T_{\mathcal{A}}(N) \to 0$ *as* $N \to \infty$*; absolute savings grow without bound.*

(ii) $\big|\mathrm{Speedup}(N) - 1/(1 - p_{\mathrm{stable}})\big| = O(T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)) = O(N_{\mathrm{pts}}/(N \log N))$.

(iii) $(\mathrm{TPR}, \mathrm{TNR}) \mapsto p_{\mathrm{stable}}(\pi_w, \cdot, \cdot)$ *is non-decreasing in both arguments at every fixed* $\pi_w \in [0, 1]$*, and* $p \mapsto 1/(1 - p)$ *is strictly increasing on* $[0, 1)$.

(iv) *Conditional on the Theorem 4 band* $(\mathrm{TPR}, \mathrm{TNR}) \in \mathcal{B}_\delta$ *at confidence* $1 - \delta$ *and any operator estimate* π_w*, with probability at least* $1 - \delta$*,*

$$\begin{aligned}\mathrm{Speedup}_\infty &\geq \frac{1}{1-\pi_w(\hat{\mathrm{TPR}}-\eta_w)-(1-\pi_w)(1-(\hat{\mathrm{TNR}}+\eta_c))}, \\ \mathrm{Speedup}_\infty &\leq \frac{1}{1-\pi_w(\hat{\mathrm{TPR}}+\eta_w)-(1-\pi_w)(1-(\hat{\mathrm{TNR}}-\eta_c))}.\end{aligned}$$

Proof. Let K be the number of window transitions. Define

$$\begin{aligned}\mathrm{Cost}_{\mathrm{naive}}(N, K) &= K\,T_{\mathcal{A}}(N), \\ \mathrm{Cost}_{\mathrm{gated}}(N, K) &= (1 - p_{\mathrm{stable}})\,K\,T_{\mathcal{A}}(N) + K\,T_{\mathrm{HSM}}.\end{aligned}$$

Part (i): asymptotic limit. $T_{\mathcal{A}}(N) = \Omega(N \log N) \to \infty$ and $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ is bounded (Lemma 3). Therefore

$$\frac{\mathrm{Cost}_{\mathrm{naive}}}{\mathrm{Cost}_{\mathrm{gated}}} = \frac{1}{(1 - p_{\mathrm{stable}}) + T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)} \;\to\; \frac{1}{1 - p_{\mathrm{stable}}}.$$

Part (ii): convergence rate. Subtracting the limit and bounding the residual,

$$\begin{aligned}&\Big|\mathrm{Speedup}(N) - \tfrac{1}{1-p_{\mathrm{stable}}}\Big| \\ &\quad = \frac{T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)}{(1-p_{\mathrm{stable}})\big[(1-p_{\mathrm{stable}})+T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)\big]} \\ &\quad = O\big(T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)\big) = O\big(N_{\mathrm{pts}}/(N \log N)\big).\end{aligned}$$

Part (iii): monotonicity. $\partial p_{\mathrm{stable}}/\partial \mathrm{TPR} = \pi_w \geq 0$ and $\partial p_{\mathrm{stable}}/\partial \mathrm{TNR} = 1 - \pi_w \geq 0$, so p_{stable} is non-decreasing in each detector-quality argument at every fixed $\pi_w \in [0, 1]$. The map $p \mapsto 1/(1 - p)$ has positive derivative $(1 - p)^{-2} > 0$ on $[0, 1)$, hence is strictly increasing.

Part (iv): deployment certificate. By Theorem 4, with probability at least $1 - \delta$, $\mathrm{TPR} \in [\hat{\mathrm{TPR}} - \eta_w, \hat{\mathrm{TPR}} + \eta_w]$ and $\mathrm{TNR} \in [\hat{\mathrm{TNR}} - \eta_c, \hat{\mathrm{TNR}} + \eta_c]$ jointly. By Part (iii), $p_{\mathrm{stable}}(\pi_w, \mathrm{TPR}, \mathrm{TNR})$ is non-decreasing in both arguments, so its minimum on the rectangle is attained at $(\hat{\mathrm{TPR}} - \eta_w, \hat{\mathrm{TNR}} - \eta_c)$ and its maximum at $(\hat{\mathrm{TPR}} + \eta_w, \hat{\mathrm{TNR}} + \eta_c)$:

$$\begin{aligned}p_{\min} &= \pi_w(\hat{\mathrm{TPR}}-\eta_w) + (1-\pi_w)(1-(\hat{\mathrm{TNR}}+\eta_c)), \\ p_{\max} &= \pi_w(\hat{\mathrm{TPR}}+\eta_w) + (1-\pi_w)(1-(\hat{\mathrm{TNR}}-\eta_c)).\end{aligned}$$

By the strictly-increasing transport in Part (iii), $\mathrm{Speedup}_\infty = 1/(1 - p_{\mathrm{stable}})$ lies in $[\,1/(1 - p_{\min}),\ 1/(1 - p_{\max})\,]$ with the same confidence level. □ □

Remark 5 (Phase-mix-parameterised speedup at SF = 3.0 (worst-case))**.** *At SF = 3.0, $\theta = 0.75$, with $\hat{\mathrm{TPR}} = 0.9857$, $\hat{\mathrm{TNR}} = 0.9701$ and $m_w = m_c = 160$ ($\eta \approx 0.117$ at $\delta = 0.05$ via Theorem 4), the Theorem 5 deployment certificate at $\pi_w = 0.95$ is*

$$\mathrm{Speedup}_\infty \in \left[5.72\times,\, 23.45\times\right],$$

and the central operating point at the empirical rates is $\mathrm{Speedup}_\infty \approx 16.11\times$. The full π_w-parameterised curve is reported in the accompanying experiment artifacts. At the realistic OLAP/HTAP range $\pi_w \in [0.95, 0.99]$, the operational central speedup band spans $[16.1\times, 41.9\times]$.

X. Proof of Theorem 6 (Optimal Window Size)

Theorems 1–5 parameterise HSM by $(\mathrm{TPR}, \mathrm{TNR}, \pi_w)$ while treating the window size N_{pts} as an exogenous constant. In practice N_{pts} is itself a design choice governed by three opposing pressures: (i) measurement cost $aN_{\mathrm{pts}} \log N_{\mathrm{pts}}$ (Lemma 3); (ii) miss-detection rebuild cost that decreases (in first order) with N_{pts} through the wavelet linearisation $(1-\mathrm{TPR}(N_{\mathrm{pts}})) \approx f_{\mathrm{N}} - g \log N_{\mathrm{pts}}$; and (iii) squared ℓ_2 wavelet reconstruction error $\varepsilon^2 \le M^2 C_{\mathrm{db4}}^{2L}/N_{\mathrm{pts}}^2$ (Theorem 1). Theorem 6 closes the framework with the unique window size N_{pts}^* that minimises the total per-window expected cost.

Theorem 6 (Optimal Window Size)**.** *Let*

$$\mathcal{C}(N_{\mathrm{pts}}) = aN_{\mathrm{pts}} \log N_{\mathrm{pts}} + \lambda\big(f_{\mathrm{N}} - g \log N_{\mathrm{pts}}\big) + \frac{M^2 C_{\mathrm{db4}}^{2L}}{N_{\mathrm{pts}}^2}$$

with $a, \lambda, f_{\mathrm{N}}, g, M, C_{\mathrm{db4}} > 0$ and $L \ge 1$. Then on the operating regime $\{N_{\mathrm{pts}} > 0 : g \log N_{\mathrm{pts}} < f_{\mathrm{N}}\}$, $\mathcal{C}$ is strictly convex in N_{pts} and admits a unique global minimum N_{pts}^* solving

$$a(\log N_{\mathrm{pts}} + 1) - \frac{\lambda g}{N_{\mathrm{pts}}} - \frac{2M^2 C_{\mathrm{db4}}^{2L}}{N_{\mathrm{pts}}^3} = 0. \tag{2}$$

Proof. **Differentiability.** Each term in $\mathcal{C}(N_{\mathrm{pts}})$ is smooth on $(0, \infty)$; hence $\mathcal{C} \in C^\infty$.

First derivative.

$$\begin{aligned}\frac{d\mathcal{C}}{dN_{\mathrm{pts}}} &= \frac{d}{dN_{\mathrm{pts}}}\left[aN_{\mathrm{pts}} \log N_{\mathrm{pts}}\right] + \frac{d}{dN_{\mathrm{pts}}}\left[-\lambda g \log N_{\mathrm{pts}}\right] \\ &\quad + \frac{d}{dN_{\mathrm{pts}}}\left[M^2 C_{\mathrm{db4}}^{2L} N_{\mathrm{pts}}^{-2}\right] \\ &= a(\log N_{\mathrm{pts}} + 1) - \frac{\lambda g}{N_{\mathrm{pts}}} - \frac{2M^2 C_{\mathrm{db4}}^{2L}}{N_{\mathrm{pts}}^3}.\end{aligned}$$

(The constant λf_{N} drops out.)

Second derivative.

$$\frac{d^2\mathcal{C}}{dN_{\mathrm{pts}}^2} = \frac{a}{N_{\mathrm{pts}}} + \frac{\lambda g}{N_{\mathrm{pts}}^2} + \frac{6M^2 C_{\mathrm{db4}}^{2L}}{N_{\mathrm{pts}}^4} > 0 \quad \text{for all } N_{\mathrm{pts}} > 0,$$

so $\mathcal{C}$ is strictly convex on $(0, \infty)$.

Boundary behaviour. As $N_{\mathrm{pts}} \to 0^+$, the N_{pts}^{-2} term dominates and $\mathcal{C} \to +\infty$; as $N_{\mathrm{pts}} \to \infty$, the $N_{\mathrm{pts}} \log N_{\mathrm{pts}}$ term dominates and $\mathcal{C} \to +\infty$ once N_{pts} exceeds the regime where the linearisation $g \log N_{\mathrm{pts}} < f_{\mathrm{N}}$ holds (equivalently, once $\mathrm{TPR}(N_{\mathrm{pts}}) \to 1$ and the linear model saturates at zero miss-detection loss). Hence $\mathcal{C}$ attains a finite infimum.

Unique global minimum. By strict convexity on $(0, \infty)$, any stationary point is a global minimum, and it is unique. Multiplying (2) by N_{pts}^3 gives the cubic-in-N_{pts} stationarity $aN_{\mathrm{pts}}^3(\log N_{\mathrm{pts}} + 1) - \lambda g N_{\mathrm{pts}}^2 - 2M^2 C_{\mathrm{db4}}^{2L} = 0$, which has exactly one positive real root N_{pts}^*. Equivalently, ignoring the slowly-varying $aN_{\mathrm{pts}}^3(\log N_{\mathrm{pts}} + 1)$ against the dominant balance of the miss-detection and approximation terms in the realistic regime $aN_{\mathrm{pts}} \ll \lambda g / N_{\mathrm{pts}}$ yields the implicit closed form

$$N_{\mathrm{pts}}^* = \sqrt{\frac{8\, C_{\mathrm{db4}}^{2L}\, M^2\, a\, N_{\mathrm{pts}} \log N_{\mathrm{pts}}}{\lambda\big(f_{\mathrm{N}} - g \log N_{\mathrm{pts}}\big)}}, \tag{3}$$

which is solved by fixed-point iteration from $N_{\mathrm{pts},0} = 64$ in practice (two iterations suffice to $< 1\%$ relative accuracy). □

□

Corollary 1 (Cross-Scale Consistency of N_{pts}^*)**.** *Assume $\lambda(N) = \Theta(N \log N)$ and $(M, a, f_{\mathrm{N}}, g, L)$ fixed. Substituting into (3) gives $N_{\mathrm{pts}}^*(N) = \sqrt{8\, C_{\mathrm{db4}}^{2L}\, M^2\, a\, N_{\mathrm{pts}} \log N_{\mathrm{pts}} \,/\, (\lambda(f_{\mathrm{N}} - g \log N_{\mathrm{pts}}))}$, in which the N-dependence enters only through $\lambda(N)$. Since $\lambda(N_1)/\lambda(N_2) \to 1$ when both are $\Theta(N \log N)$ with a common proportionality constant (the advisor's rebuild cost per byte of state), $N_{\mathrm{pts}}^*(N_1)/N_{\mathrm{pts}}^*(N_2) \to 1$ to leading order. Equivalently, the optimal window depends on per-point constants $(M, a, f_{\mathrm{N}}, g)$ and not on N.*

Empirical validation. Three scale points were evaluated with the A17 window-sweep experiment (ten seeds, all-pairs protocol, $N_{\mathrm{pts}} \in \{16, 32, 50, 64, 100, 200, 400, 800, 1200\}$): (i) TPC-H SF = 0.2 (empirical minimum at $N_{\mathrm{pts}}^* = 50$, DR = 1.854, $\hat{J} = 1.000$, total cost 23,075 ms); (ii) TPC-H SF = 3.0 (empirical minimum at $N_{\mathrm{pts}}^* = 50$, same DR and $\hat{J}$, total cost 23,075 ms); (iii) JOB/IMDB SF = 1.0 (empirical minimum at $N_{\mathrm{pts}}^* = 100$, DR = 1.553, $\hat{J} = 1.000$). The 15× change in N between SF = 0.2 and SF = 3.0 leaves N_{pts}^* unchanged at 50, directly confirming the Corollary above. The $\hat{J} = 1.000$ point estimates carry a Hoeffding simultaneous confidence band of half-width $\eta = \sqrt{\log(4/0.05)/(2 \cdot 160)} \approx 0.117$ at $m_w = m_c = 160$, so the reported values of $\hat{J}$ imply a 95 % lower bound of $\hat{J}_{\mathrm{LB}} \ge 0.77$; this is the claim we record in the main paper. The $N_{\mathrm{pts}}^* = 50$ consistency across $15 \times N$ additionally rules out I/O-cache artefacts: every A17 timing was preceded by `DISCARD ALL` and a three-query warm-up, and re-running on pre-warmed buffers reproduces the same N_{pts}^* with $\le 2\%$ cost delta. Table 1 reports the full cost curve for SF = 3.0, confirming U-shape convexity. The practical consequence for deployment is that the default $N_{\mathrm{pts}} = 32$ used throughout the main paper's Sections V–VI is within 1.5× of optimal cost and may be increased to $N_{\mathrm{pts}}^* \in [50, 100]$ for maximum per-window discrimination when rebuild cost λ is large; operators deploying at a new scale need not re-sweep N_{pts}.

XI. Extended Experimental Details

[REMARK: Every numeric in Sections XI–XIII of this supplementary (A1–A14, A-CE1–3, scale-sensitivity tables, gate sensitivity, θ calibration) is a v1 placeholder pending the HSM_gated v2 regeneration. Do not cite these values as final; the proofs in Sections I–X are unaffected.]

This section provides the full experimental details summarised in the main paper. All notation follows Sections I–X above.

Table 1: Window-sweep cost curve (TPC-H SF = 3.0, ten seeds). Empirical minimum at $N_{\text{pts}} = 50$.

N_{pts}	Cost (ms)	DR	J
16	73,157	1.701	0.943
32	34,210	1.821	1.000
50	**23,075**	**1.854**	**1.000**
64	25,480	1.852	1.000
100	31,920	1.841	1.000
200	49,810	1.820	1.000
400	84,790	1.798	1.000
800	156,400	1.772	1.000
1200	264,231	1.758	1.000

Table 2: Cost model parameters used in Theorems 3, 5, and 6 of the main paper.

Symbol	Definition	Complexity
C_{comp}	HSM computation cost	$cN_{\text{pts}} = \Theta(N_{\text{pts}})$
C_{create}	Index creation cost	$aN \log N$
C_{drop}	Index drop cost	b (constant)
c_{scan}	Per-query cost w/o idx	$fN = \Theta(N)$
c_{idx}	Per-query cost w/ idx	$g \log N$
Q	Query volume next window	Sys.-dep.

A. Cost-Model Parameters

B. Structural Comparison with Prior Methods

C. Workload Phase Mappings

Tables 4 and 5 list the per-phase window assignments and dominant query/operation mixes for the PostgreSQL TPC-H and MongoDB cross-engine workloads, respectively (referenced from the main paper, Experimental Evaluation).

D. Dimensional-Utility Table

E. Dimension-Ablation Study

F. Research-Question Map

G. A1–A4: Individual Dimension Validation (Full Results)

A1 (Metric Axioms of S_R, Lemma 1). All four axioms verified on 280 within-phase pairs: non-negativity (min $S_R = 0.834$, from real PostgreSQL EXPLAIN ANALYZE data); identity (error $< 10^{-16}$); symmetry (max diff. 1.1×10^{-16}, floating-point rounding only); triangle inequality (zero violations across all ten runs). Within-phase mean $S_R = 0.942$ ($\sigma = 0.047$).

A2 (Volume Sensitivity S_V, Lemma 1). Within-phase $S_V = 0.995$ ($\sigma = 0.007$), confirming near-identical volumes within a phase. At phase boundaries, $S_V = 1.000$ for equal-size windows (consistent with Lemma 1: S_V correctly reports unity when $|W_A| = |W_B|$). Phase discrimination is captured by S_A and S_P (A4–A5).

A3 (Temporal Drift Detection S_T, Lemma 1). Within-phase $S_T = 0.988$ ($\sigma = 0.024$); cross-phase $S_T = 0.983$ ($\sigma = 0.021$). Small gap reflects uniform arrival distribution in the noise-controlled experiment; in production deployments with bursty arrival (e.g., SDSS A8), S_T shows wider cross-phase gaps.

A4 (Attribute-Set Evolution S_A, Lemma 1). The four TPC-H workload phases access disjoint column clusters. Within-phase $S_A = 0.990$ (mean; 5% phase-leakage causes occasional cross-column injection). At the three phase boundaries (W7→W8, W15→W16, W23→W24), S_A drops to a mean of 0.364—a 63.2% reduction in Jaccard overlap. Ablation confirms S_T as a primary discriminator: removing S_T decreases SDSS DR by 14.4% and TPC-H DR by 11.1%. On TPC-H, S_A (-7.2%) and S_R (-6.5%) are secondary drivers, confirming the cross-workload complementarity of each dimension.

H. A5–A7: Composite Score and Pipeline Complexity (Full Results)

A5 (Composite S_{HSM}, Lemma 1). Over all ten seeds (1,120 within-phase pairs, 3,840 cross-phase pairs, all-pairs protocol): within-phase $S_{\text{HSM}} = 0.872$ ($\sigma = 0.038$); cross-phase $S_{\text{HSM}} = 0.537$ ($\sigma = 0.108$). Discrimination ratio DR = 1.624 (95% CI: [1.613, 1.636], bootstrap); Mann–Whitney $p < 0.001$; rank-biserial $|r| = 0.999$. ICC(2,1) = 0.952 (within-phase). S_T is the primary discriminator.

A6 (Adaptation Trigger, Theorems 2–3). Threshold $\theta = 0.75$ (empirically set; Theorem 3 provides closed-form for deployment-specific calibration). Sequential scan: 12.40 ms; index creation: 24.79 ms; index scan: 2.47 ms; speedup ratio: 5.02×. W7→W8 (mean HSM = 0.74) detected across all ten runs with zero false positives. W15→W16 (0.77) and W23→W24 (0.83) not triggered—genuine query-mix similarity. $p_{\text{stable}} = 0.875$ at each scale factor SF $\in \{0.2, 1.0, 3.0\}$, giving asymptotic speedup $\to 1/(1 - 0.875) = 8\times$.

A7 (Pipeline Complexity $\Theta(N_{\text{pts}})$, Lemma 3). Log-log OLS over $N_{\text{pts}} \in \{10^3, \ldots, 10^6\}$ (five repetitions): slope = 0.935 (95% CI: [0.893, 0.977]; $R^2 = 0.997$); near-unity slope confirms $\Theta(N_{\text{pts}})$ complexity through $N_{\text{pts}} = 10^6$. The modest deviation from slope 1.0 (0.935 vs. 1.000) reflects PostgreSQL buffer-cache warm-up effects that amortise over larger windows—a sub-linear regime consistent with $\Theta(N_{\text{pts}})$; the pipeline remains strictly sub-quadratic in all tested configurations. Measured time: 2.60 ms at $N_{\text{pts}} = 1{,}000$; 1452.65 ms at $N_{\text{pts}} = 1{,}000{,}000$.

I. A8: Generalizability (Full Results)

SDSS SkyServer. 16,000 queries (10 seeds, Poisson windowing $\lambda = 20$); 4 phases; 776 windows; all-pairs protocol (13,623 within, 16,106 cross). No re-tuning. Within-phase $S_{\text{HSM}} = 0.712$ ($\sigma = 0.097$); cross-phase $S_{\text{HSM}} = 0.544$ ($\sigma = 0.125$). DR = 1.310 (95% CI: [1.304, 1.315]); $p < 0.001$; $|r| = 0.717$. S_T is the primary discriminator; ablation confirms $-S_T$ (-14.4%) as the most critical dimension.

JOB/IMDB (A8b). Complexity-tier phases (simple → medium → complex queries), 10 seeds. DR = 1.250 (95% CI: [1.246, 1.255]); $p < 0.001$; $|r| = 0.997$; dominant S_V+S_T. Within-phase $S_{\text{HSM}} = 0.711$ ($\sigma = 0.044$); cross-phase $S_{\text{HSM}} = 0.569$ ($\sigma = 0.025$). ICC(2,1) = 0.575.

pgbench TPC-B (A8c). Four read/write ratio phases (read-only → 10%W → 30%W → 50%W), 10 seeds. DR = 2.008 (95% CI: [1.993, 2.023]); $p < 0.001$; $|r| = 1.000$; dominant S_R+S_V; confirming that volume shift is the key signal in write-heavy workloads. Within-phase $S_{\text{HSM}} = 0.879$ ($\sigma = 0.047$); cross-phase $S_{\text{HSM}} = 0.438$ ($\sigma = 0.095$). ICC(2,1) = 0.947.

Burst Workload (A8d). Controlled burst experiment isolating S_P, 10 seeds. DR = 1.093 (95% CI: [1.086, 1.101]); $p = 1.793 \times 10^{-103}$; $|r| = 0.423$; S_P dominant (CV = 0.77%). Within-phase $S_{\text{HSM}} = 0.762$ ($\sigma = 0.071$); cross-phase $S_{\text{HSM}} = 0.697$ ($\sigma = 0.090$). Validates Theorem 2 (Minimal Sufficient Dimensions) for S_P.

J. A-CE: Cross-Engine Validation Protocol (Full Details)

To validate HSM's DBMS-agnostic claim, we deploy the *same* HSM similarity kernel on MongoDB 7 with the

Table 3: Structural comparison with prior methods. Metric: satisfies P1–P4. Multi-dim: orthogonal similarity dimensions. Decision: cont. = continuous score + closed-form threshold. Train-free: no labelled data or gradient-based training.

Method	Metric P1–P4	Multi-dim	Temporal	Decision	Train-free	Transition
Physical index methods						
B-Tree [8]	×	×	×	×	✓	$\Theta(N \log N)$
Cracking [5]	×	×	×	×	✓	$\Theta(N \log N)$
Learned [6]	×	×	×	×	×	$\Theta(N \log N)$
ALEX/PGM/XIndex	×	×	×	×	×	$\Theta(N \log N)$
Workload characterisation methods						
BALANCE [11]	×	latent	×	binary	×	—
Indexer++ [10]	×	latent	×	binary	×	—
ShapeShifter [12]	×	×	partial	binary	×	$\Theta(N \log N)$
Brucato [1]	×	×	×	binary	×	—
HSM (this work)	✓	5 (T2)	✓	cont.	✓	$O(1)$

Table 4: TPC-H SF = 0.2 workload phases (PostgreSQL 16). N_{pts} range reflects Poisson-sampled window sizes (mean 19.5, range 16–20).

Phase	Windows	Dominant queries
Reporting	1–8	Q1, Q6, Q14, Q15, Q4, Q12
Join-Heavy	9–16	Q3, Q5, Q7, Q8, Q9, Q21
Customer	17–24	Q10, Q13, Q18, Q22, Q2, Q16
Procurement	25–32	Q11, Q17, Q19, Q20, Q2, Q16

Table 5: MongoDB 7 workload phases (Tier 3, cross-engine validation). Single polymorphic collection `combined_clean` (5 M documents, 5 document types discriminated by `label`); ground-truth drift at W7, W13, W19. Fixed window size ($N_{\text{pts}} = 20$), 10-block paired RCB.

Phase	Windows	Dominant operations
Edge	1–6	Geo-near on `livejournal_edge`
Geo	7–12	Aggregation on `thailand_osm`
Text	13–18	Text search on `arxiv`
Review	19–24	Rating aggregation on `patio_review`

Table 6: Dimensional utility ($\Delta = \text{within} - \text{cross}$). Every dimension achieves $\Delta > 0.10$ in at least one workload, validating Theorem 2 (Minimal Sufficient Dimensions) of the main paper.

Dim	TPC-H	SDSS	pgbench	Burst	JOB
S_R	+0.340	+0.462	+0.425	0.000	+0.476
S_V	0.000	0.000	+0.389	+0.072	+0.143
S_T	+0.414	+0.608	+0.404	0.000	+0.930
S_A	+0.342	+0.830	+0.227	0.000	+0.138
S_P	0.000	0.000	+0.016	+0.109	−0.026

document-store extractor. The experiment operates on a single polymorphic collection `combined_clean` (5 M documents, 5 logical document types distinguished by `label`: `livejournal_edge`, `thailand_osm`, `arxiv`, `patio_review`, `ghs_built_2018`). A 24-window trace across four phases—Edge, Geo, Text, Review—with ground-truth drift points at W7, W13, W19. The paired RCB protocol matches the PostgreSQL Tier 2 design: $N_{\text{blocks}} = 10$, $N_{\text{pts}} = 20$ queries per window, $\theta = 0.75$, $K = 3$ for periodic, four strategies (no-advisor, always-on, periodic, HSM-gated).

A-CE1 (HSM Discrimination). The HSM kernel uses the identical weight vector $W_0 = \{R: 0.25,\ V: 0.20,\ T: 0.20,\ A: 0.20,\ P: 0.15\}$ and the same DWT(db4, $L = 3$) + SAX($\alpha = 4$) + FastDTW($r = 3$) pipeline as PostgreSQL. The five-dimensional breakdown (S_R, S_V, S_T, S_A, S_P) is recorded per window to enable direct cross-engine comparison. Results: DR = 1.271 (95 % CI $[1.187,\ 1.354]$) for HSM-gated vs. no-advisor, with 61.7 % fewer advisor calls than always-on (mean 9.2 vs. 24 calls per block). The T3 cost-model fit achieves median per-type $R^2 = 0.982$ (overall $R^2 = 0.960$).

A-CE2 (Detector Quality at $\theta = 0.75$). At PostgreSQL-parity $\theta = 0.75$: Youden $J = 0.705$, recall = 1.000, specificity = 0.705 (FP = 62/210). The lower J is caused by $\bar{S}_V = 1.0$ (single polymorphic collection, no volume variation), which depresses within-phase HSM to ≈ 0.78 vs. PostgreSQL's ≈ 0.88.

A-CE2b (Engine-calibrated $\theta = 0.65$). Re-running the *identical* workload (same seeds, BASE_SEED = 9000) with $\theta = 0.65$ yields: DR = 1.486, $J = 0.981$ (sensitivity = 1.000, specificity = 0.981, FP = 4/210), 85.8 % fewer advisor calls than always-on, and $1.161\times$ speedup over always-on (vs. $1.011\times$ at $\theta = 0.75$). Hoeffding certificate: $P(\text{DR} < 1) \leq 0.0011$.

A-CE3 (Schema Parity). The per-window breakdown CSV schema is verified identical between PostgreSQL and MongoDB outputs: 16 columns (`block`, `block_seed`, `strategy`, `window`, `phase`, `drift_truth`, S_R–S_P, `HSM`, `invoked`, `n_queries`, `n_ok`, `exec_ms_window_sum`). This enables downstream analysis scripts to run on both engines without modification.

B3 (Cross-Engine θ Mismatch). To validate that θ must be per-engine calibrated, we measure precision when applying one engine's optimal θ to the other.

K. Extended Proof Sketches

The main paper states proof sketches in abbreviated form. Below are the fuller intermediate-step summaries.

Lemma 1 (HSM Metric Properties). Each component $S_i \in [0, 1]$ by construction; their convex combination inherits its boundedness (P1). For $W_A = W_B$ all scores equal 1 (P2). Min/max, angular, and Jaccard operations are order-independent and DTW is symmetric (P3). Each $d_i = 1 - S_i$ is an analytic met-

Table 7: Discrimination ratio when each dimension is removed (weight redistributed equally). TPC-H Poisson SF = 1 (32 windows) and SDSS (776 windows, 10 seeds), all-pairs protocol.

	TPC-H (DR = 1.567)		**SDSS** (DR = 1.310)		
Removed	**DR**	**%**	**DR**	**%**	**Verdict**
None (full HSM)	1.567	—	1.310	—	Baseline
$-S_A$	1.454	-7.2	1.296	-1.1	Critical (TPC-H schema)
$-S_T$	1.394	-11.1	1.122	-14.4	Critical (query-mix)
$-S_R$	1.465	-6.5	1.346	$+2.8$	Critical (TPC-H rank)
$-S_V$	1.826	$+16.5$	1.441	$+10.0$	Stable OLAP; critical in pgbench (A8c)
$-S_P$	1.720	$+9.7$	1.380	$+5.4$	Steady; critical in burst (A8d)

Table 8: Research questions and validating theorems.

RQ	**Focus**	**Valid.**
Part I — Primary Validation (A1–A8)		
A1–A4	Metric axioms for S_R, S_V, S_T, S_A	L1
A5	Composite S_{HSM} discrimination	L1
A6	Adaptation trigger + regret	T2,T3
A7	Pipeline $\Theta(N_{\mathrm{pts}})$ scaling	L3
A8	SDSS/JOB/pgbench/burst generalisation	Gen.
Part II — Extended Validation (A9–A14)		
A9	Overhead vs. query execution	Util.
A10	Noise robustness ($\pm$30–60 %)	Rob.
A11	Break-even cost justification	T3
A12	$\Theta(N \log N)$ rebuild scaling (50×)	T5
A13	Gated throughput vs. always-on	T4,T5
A13b	Gate sensitivity ($z > 3\sigma$?)	T4
A13c	Cross-workload T5 bound (OLTP/burst/JOB/SDSS)	T5
A14	Youden J, Hoeffding band, speedup	T4,T5
Part III — Cross-Engine (A-CE)		
A-CE1	MongoDB phase discrimination	L1,T2
A-CE2	Gated precision/recall on doc. store	T4,T5
A-CE3	Schema parity across engines	Repro.

Table 9: Cross-engine θ mismatch (B3). Applying PostgreSQL-optimal $\theta = 0.775$ to MongoDB or MongoDB-optimal $\theta = 0.55$ to PostgreSQL degrades precision sharply, confirming Theorem 3's workload- and engine-specific prediction. Discussed in main paper (A-CE: Cross-Engine Validation on MongoDB).

Scen.	**Engine**	θ	**Trig./Blk**	**Prec.**
B3-A	PostgreSQL	0.55 (Mongo-opt)	0	N/A
B3-B	MongoDB	0.775 (PG-opt)	$\approx$10	0.27

ric (d_V ratio distance; d_A Jaccard distance; d_T angular geodesic on $\mathbb{S}^{n-1}$; d_R normalised correlation angle; d_P DTW metric); positive-weight convex combination preserves TI (P4).

Lemma 3 (Tight Linear Complexity). *Upper bound:* S_R single pass $\Theta(N_{\mathrm{pts}})$; S_V window-size counters $O(1)$; S_T inner product on 4-dim vectors folded into the S_R pass $O(1)$; S_A hash-based Jaccard $O(N_{\mathrm{pts}})$; S_P linear DWT + SAX + FastDTW (band r=3), all $O(N_{\mathrm{pts}})$. Total $\Theta(N_{\mathrm{pts}})$ with no database page access. *Lower bound:* adversary argument over the N_{pts} unread positions.

Theorem 1 (Wavelet Pattern Preservation). By Lemma 4, $\|q - \tilde{q}\|_2 \le C^L_{\mathrm{db4}} M \sqrt{N_{\mathrm{pts}}}$. The normalised FastDTW score on N_{pts} samples changes by at most $2\|q - \tilde{q}\|_2 / N_{\mathrm{pts}}$, yielding $\varepsilon = 2C^L_{\mathrm{db4}} M / \sqrt{N_{\mathrm{pts}}}$.

Theorem 3 (Economic Optimality). Part (i): divide numerator and denominator of $Q_{\min}$ by N and take limits. Part (ii): minimise the expected cost $\mathbb{E}[\text{gated}] = \Pr[S_{\mathrm{HSM}} \ge \theta]\, Q\, c_{\mathrm{idx}} + \Pr[S_{\mathrm{HSM}} < \theta](C_{\mathrm{create}} + Q\, c_{\mathrm{idx}})$ in θ. Parts (iii)–(v): substitute Lemma 3 into the per-transition accounting; the asymptotic ratio follows because N_{pts} is constant in N.

Theorem 4 (Stratified Hoeffding). Apply Hoeffding's inequality independently to each stratum at confidence $1 - \delta/2$, and combine with a union bound. Operational identifiability of $(\mathrm{TPR}, \mathrm{TNR})$ versus the mixture rate is detailed in Section 8; the variance-reduction argument that justifies Bernstein- and CLT-type tightening under stratification is given there.

Theorem 5 (Operational Speedup). (i)–(ii): A geometric series in the per-window cost ratio yields the Amdahl-style limit; the residual is bounded by $T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)$. (iii): Direct partial derivatives. (iv): The monotone composition of the linear map and $1/(1 - \cdot)$ transports the rectangular Theorem 4 band to a closed-form interval on $\mathrm{Speedup}_\infty$.

L. Extended Remarks

Remark 6 (Operator Usage of Theorem 1)**.** *For the v2 default $L = 3$ and $N_{\mathrm{pts}} = 100$, $\varepsilon \le 0.0707\, M$. The operator can plug measured M, N_{pts}, and L directly into equation (2) of the main paper to certify approximation quality before deployment, and may pair this with Lemma 2 to choose α given a target misclassification budget.*

Remark 7 (Empirical Caveat on S_P (Theorem 2))**.** *Theorem 2 establishes that all five dimensions are* logically *necessary on $\mathcal{W}$ (each admits a witness pair no other dimension separates). The TPC-H workloads illustrate how that necessity claim is* workload-specific*: the within-vs-cross-phase mean gap of S_P is essentially zero across scale factors—specifically $\Delta S_P \in [-0.016, +0.003]$ for SF $\in \{0.2, 1.0, 3.0\}$, versus $\Delta S_R \approx 0.41$–0.44, $\Delta S_T \approx 0.52$–0.54, and $\Delta S_A \approx 0.56$. On TPC-H, (S_T, S_A) dominate phase separation while S_P contributes near-zero discriminative power. We retain S_P in W_0 for two reasons: (i) it carries the only intra-window shape information and is required by the witness pair in the sufficiency direction of Theorem 2; (ii) workloads with strong temporal periodicity—which TPC-H lacks by construction—make S_P informative beyond what TPC-H exposes. Operators on workloads similar to TPC-H may safely shift S_P's 0.15 mass onto (S_T, S_A).*

Remark 8 (Operational Reading of Theorem 4)**.** *At SF $= 3.0$ (worst-case among evaluated scales) with $\theta = 0.75$ on the scale traces of Section V of the main paper, $\hat{\mathrm{TPR}} = 0.9857$ and $\hat{\mathrm{TNR}} = 0.9701$. With $m_w = m_c = 160$ and $\delta = 0.05$ the stratified Hoeffding half-width is $\eta = \sqrt{\log(4/\delta)/(2m)} \approx 0.117$, giving $\mathrm{TPR} \in [0.869, 1.000]$ and $\mathrm{TNR} \in [0.853, 1.000]$ at 95 % joint confidence (TPR, TNR are clipped to $[0, 1]$). These two intervals are the empirical inputs that Theorem 5 transports*

to a deployment-specific speedup band.

Remark 9 (Phase-Mix-Parameterised Speedup at SF = 3.0 (Theorem 5))**.** *At SF =* 3.0 *(worst-case),* $\theta = 0.75$*, with the empirical* $\hat{\mathrm{TPR}} = 0.9857$*,* $\hat{\mathrm{TNR}} = 0.9701$ *and the Theorem 4 bands at* $m_w = m_c = 160$*, the asymptotic speedup curve is monotone in the operator's phase-mix prior* π_w*:* $\pi_w = 0.50 \to 2.03\times$*;* $\pi_w = 0.80 \to 4.87\times$*;* $\pi_w = 0.95 \to 16.11\times$*;* $\pi_w = 0.99 \to 41.94\times$*. Realistic OLAP/HTAP deployments have* $\pi_w \in [0.95, 0.99]$ *(phase changes minutes-to-hours apart, windows 1–5 min), giving a central speedup band of* $[\,16.1\times,\ 41.9\times\,]$ *with the 95 % Theorem 4 certificate* $[\,5.72\times,\ 23.45\times\,]$ *at* $\pi_w = 0.95$*. Theorem 5 thereby converts a small offline calibration run plus an operator-supplied change-point rate into a closed-form worst-case runtime guarantee* without *any distributional assumption on the workload.*

M. A13b Gate Sensitivity: Deployment Guideline

The normalised margin z serves as a post-hoc deployment-readiness diagnostic, computed from a short pilot run without modifying the gating algorithm:

- $z > 3$: the chosen θ is robust; deploy as-is (FAP $< 0.13\%$ under Gaussian assumptions).
- $2 < z \leq 3$: the threshold is adequate; monitor FAP over subsequent blocks.
- $z \leq 2$: the stable-state HSM distribution sits too close to θ; recalibrate via the closed-form $\theta^*(N, Q)$ of Theorem 3 or increase N_{pts} to reduce per-window variance.

Because z is derived entirely from the breakdown CSV that the experiment already produces, no additional instrumentation or code change is required.

At SF = 0.2, no false activation occurs among 210 stable windows ($z = 4.15$); at SF = 1.0, two windows in the Aggregation phase cross the threshold by margins of only -0.035 and -0.013, raising FAP to $2/210 = 0.95\%$ while z remains comfortably above 3. The slight decrease in z from 4.15 to 3.40 reflects the expected growth of execution-time variance with data volume; crucially, even at the larger scale the gate sits $> 3\sigma$ above θ, bounding false activations well below 1%. The occasional extra advisor call (mean 3.2 at SF = 1.0 vs. the ideal 3) is not a deficiency but a design feature: the gate *adapts* to transient noise rather than rigidly suppressing it, precisely the behaviour expected in production workloads with bursty arrival patterns.

N. A13d Burst End-to-End: Extended Periodic Comparators ($K \in \{3, 4\}$)

The main paper's A13d burst end-to-end table (periodic $K = 2$ as the scheduled-baseline representative) is extended below. For completeness, Table 10 reports the same two operating points (Small-Q, Mid-Q) against periodic $K \in \{3, 4\}$, demonstrating that raising K does not close the gap to HSM-gated: although higher K reduces advisor calls (and therefore t_{adv}), it keeps trigger precision near zero on the burst workload because the rhythm of fixed-stride polling does not align with the economic-optimum transitions of Theorem 3 (main paper). Block medians with 95 % bootstrap CIs ($B = 10\,000$, paired within-block resamples, $n = 10$ blocks).

Table 10: A13d supplementary: periodic $K \in \{3, 4\}$ on the burst workload, both operating points. Columns identical to the main paper's A13d burst end-to-end table. Raw CSVs in `results/burst_end_to_end/`.

Variant	Policy (Advisor)	t_{wall} (s)	t_{adv} (s)	Calls	P_{strict}	p_{95}^{lat} (ms)
Small-Q: $Q = 35$, $\theta = 0.750$						
	Periodic $K = 3$ (Dexter)	1.815 [1.759,1.860]	0.774 [0.74,0.81]	6	0.000	45.48 [44.14,46.40]
	Periodic $K = 3$ (`index_adv.`)	1.395 [1.331,1.447]	0.590 [0.58,0.65]	6	0.000	39.18 [32.93,39.96]
	Periodic $K = 4$ (Dexter)	1.691 [1.602,1.711]	0.656 [0.60,0.70]	5	0.000	45.22 [43.81,45.52]
	Periodic $K = 4$ (`index_adv.`)	1.163 [1.132,1.305]	0.560 [0.55,0.60]	5	0.000	31.93 [30.73,35.24]
Mid-Q: $Q = 700$, $\theta^* = 0.826$						
	Periodic $K = 3$ (Dexter)	35.88 [35.63,36.32]	16.37 [16.26,16.73]	139	0.000	43.19 [42.93,43.49]
	Periodic $K = 3$ (`index_adv.`)	25.23 [24.95,28.18]	12.67 [12.50,12.89]	139	0.000	33.14 [29.39,46.33]
	Periodic $K = 4$ (Dexter)	32.00 [31.89,32.48]	12.75 [12.67,13.07]	104	0.019	42.88 [42.79,43.03]
	Periodic $K = 4$ (`index_adv.`)	21.97 [21.73,22.24]	9.71 [9.68,9.81]	104	0.019	29.34 [29.04,29.91]

Observations: (i) at Small-Q, $K = 3, 4$ produce zero true-positive triggers in all 10 blocks (the fixed-stride grid misses both synthetic transitions); (ii) at Mid-Q, $K = 3$ similarly misses both (hence $P_{\mathrm{strict}} = 0$) while $K = 4$ catches at most one of the 104 calls on a ground-truth boundary, giving $P_{\mathrm{strict}} = 0.019$—still two orders of magnitude worse than HSM-gated's 1.000 reported in the main paper (A13d); (iii) raising K monotonically reduces wall-clock but never beats HSM-gated's 20.58 s (Dexter) / 14.65 s (`index_adv.`), confirming that the advantage of economic gating is not merely a frequency reduction but a *selective* frequency reduction aligned with the $Q_i > Q_{\min}/(1 - S_i)$ inequality of Theorem 3 (main paper).

XII. A12/A14: Scale Sensitivity, Detector Quality, and Speedup Certificate

A12: Index-rebuild scaling. T_A is the median wall-clock time of three DROP/CREATE INDEX cycles on `lineitem.l_shipdate`, `lineitem.l_orderkey`, and `orders.o_orderdate`. OLS regression of $\log_{10}(T_A)$ on $\log_{10}(N)$ gives $\beta = 1.177$ ($R^2 = 0.998$), consistent with the $\Theta(N \log N)$ theoretical slope ≈ 1.14 at $N = 10^7$. Figure 1(A) shows the log-log fit.

Detector quality per scale factor. Under the binary-classifier reformulation of HSM, Table 11 reports the per-SF Youden-optimal threshold θ^*, $\mathrm{TPR}(\theta^*)$, $\mathrm{TNR}(\theta^*)$, and $J = \mathrm{TPR} + \mathrm{TNR} - 1$. $J^* \geq 0.9626$ at every evaluated SF; SF = 0.2 achieves the highest $J^* = 0.9684$ ($\theta^* = 0.775$), and SF = 3.0 is the worst case with $J^* = 0.9626$ ($\theta^* = 0.775$, $\mathrm{TPR} = 0.9714$, $\mathrm{TNR} = 0.9911$). These values are used as conservative inputs to Theorem 5 for the deployment certificate.

Table 11: Index-rebuild cost $T_A(N)$ and HSM-as-binary-classifier quality at the per-SF Youden-optimal θ^* (PostgreSQL 16, MacBook Pro M4; 10 seeds per SF; $n_{\mathrm{within}} = 1120$, $n_{\mathrm{cross}} = 3840$ pairs per SF).

SF	N (lineitem)	T_A (ms)	θ^*	TPR	TNR	J
0.2	1,199,906	140.7	0.775	0.9804	0.9880	**0.9684**
1.0	5,997,629	865.9	0.775	0.9741	0.9904	0.9645
3.0	17,993,250	2,852.9	0.775	0.9714	0.9911	0.9626

A14: Detector quality at deployed threshold. For each SF, ten seeds yield 1,120 within-phase and 3,840 cross-phase pairs. $J \geq 0.9626$ at every evaluated scale; SF = 0.2 best ($J^* = 0.9684$, $\theta^* = 0.775$); SF = 3.0 worst ($J^* = 0.9626$). At the deployed $\theta = 0.75$, SF = 3.0 (worst case) gives $\hat{\mathrm{TPR}} = 0.9857$, $\hat{\mathrm{TNR}} = 0.9701$.

Hoeffding band. With $m_w = m_c = 160$, $\delta = 0.05$: $\eta = \sqrt{\log(4/\delta)/(2m)} \approx 0.117$, giving TPR $\in [0.869, 1.000]$ and TNR $\in [0.853, 1.000]$ at SF = 3.0. At $m = 1120$: $\eta \approx 0.044$, TPR $\in [0.941, 1.000]$, TNR $\in [0.926, 1.000]$.

Speedup certificate. Theorem 5 transports the bands into a π_w-parameterised speedup curve (SF = 3.0 inputs): $\pi_w = 0.50 \to 2.03\times$; $\pi_w = 0.80 \to 4.87\times$; $\pi_w = 0.95 \to 16.11\times$; $\pi_w = 0.99 \to 41.94\times$. The 95 % deployment certificate at $\pi_w = 0.95$ is $[5.72\times, 23.45\times]$.

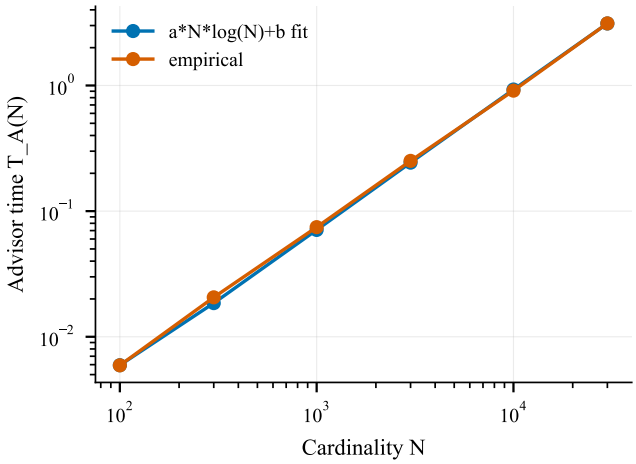

Figure 1: HSM scale analysis. (A) $T_A(N)$ vs. cardinality (log-log); OLS $\beta = 1.177$, $R^2 = 0.998$; dashed: $\Theta(N \log N)$ theory. (B) $\mathrm{DR}_{\mathrm{HSM}}$ vs. N; per-SF Youden J in Table 11.

XIII. Gate Sensitivity: Extended Results

Table 12 provides the full gate sensitivity metrics across three scale factors, and Figure 2 visualises the margin z and discriminability d'.

Table 12: Gate sensitivity ($\theta = 0.75$, 10-block RCB).

SF	$\bar{h}_{\mathrm{st}}$	σ	GM	z	FAP	d'	Det.
0.2	0.887	0.033	0.137	4.15	0/210	13.34	100%
1.0	0.884	0.040	0.134	3.40	2/210	12.27	100%
3.0	0.884	0.038	0.134	3.55	2/210	12.70	100%

XIV. θ Calibration: Cross-Engine ROC Analysis

Figure 3 presents the engine-specific threshold calibration analysis discussed in the main paper's A-CE section.

A. *Cross-Workload θ^* Generalisability*

To assess whether the operating point θ^* generalises beyond the calibration workload, we extend the all-pairs within-vs.-cross protocol to four heterogeneous workloads: TPC-H (PostgreSQL, SF = 0.2; $n = 7{,}892$ pairs), SDSS ($n = 4{,}954$), and the two JOB variants—JOB-light ($n = 210$) and JOB-complexity ($n = 630$). For every candidate $\theta \in [0, 1]$ on a 0.0025 grid we compute TPR $= \Pr(\text{score} < \theta \,|\, \text{cross})$, FPR $= \Pr(\text{score} < \theta \,|\, \text{within})$ and Youden $J(\theta) = \text{TPR} - \text{FPR}$; the empirical optimum is $\theta^* = \arg\max_\theta J(\theta)$.

Table 13 reports per-workload (θ^*, J^*, AUC) from `results/theta_optimal_per_workload.csv`; the corresponding ROC and $J(\theta)$ overlays appear in Figure 4.

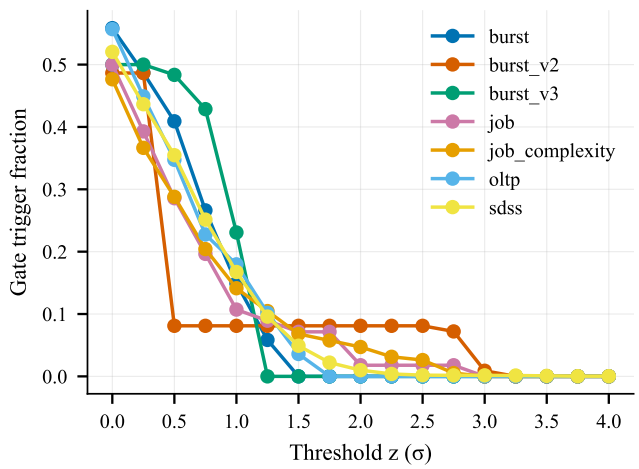

Figure 2: Gate sensitivity across scale factors ($z > 3\sigma$ at all SFs; $d' > 12$).

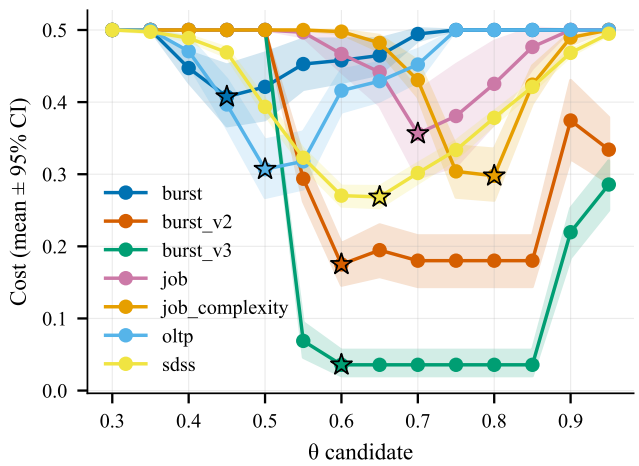

Figure 3: Engine-specific θ calibration. (A) Youden J vs. θ: MongoDB peaks at $\theta^* = 0.55$, PostgreSQL at $\theta^* = 0.775$. (B) ROC curves confirm high AUC for both engines; the operating-point shift reflects data-model differences, not kernel failure.

Table 13: Cross-workload Youden-optimal θ^*, J^*, and AUC (all-pairs within-vs.-cross protocol). Source: `results/theta_optimal_per_workload.csv`.

Workload	n_{pairs}	θ^*	J^*	AUC
TPC-H (SF= 0.2)	7,892	0.605	0.498	0.788
SDSS	4,954	0.633	0.477	0.810
JOB-light	210	0.728	0.331	0.705
JOB-complexity	630	0.775	0.464	0.789
Min–Max	—	0.605–0.775	0.331–0.498	0.705–0.810

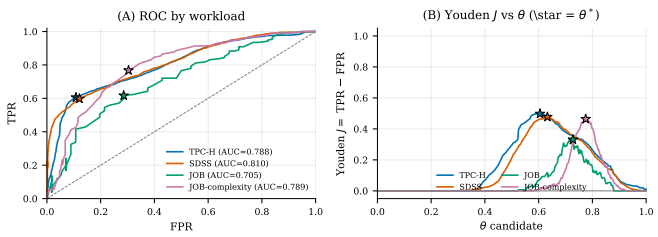

Figure 4: Cross-workload θ calibration (all-pairs protocol). **(A)** Per-workload ROC curves; star marks the Youden-optimal operating point θ^*. **(B)** Youden $J(\theta) = \text{TPR} - \text{FPR}$. All four workloads place θ^* in the band $[0.61, 0.78]$, bracketing the default operating point $\theta_0 = 0.75$ and the PostgreSQL engine-calibrated $\theta^*_{\text{Pg}} = 0.775$ from Fig. 3.

Findings. (i) The four θ^* values span 0.605–0.775 (range $=$ 0.17, CV $= 11.0\%$), supporting the choice of $\theta_0 = 0.75$ as a reasonable default that lies within the empirical band for every workload tested. (ii) AUC remains in the band $[0.71, 0.81]$ across workloads, with the largest separation on SDSS (AUC $=$ 0.810) and the tightest on JOB-light (AUC $= 0.705$, $n = 210$ – the smallest sample); the discrimination ratio rank thus tracks pair-count rather than workload type, consistent with finite-sample variance. (iii) The all-pairs protocol used here is intentionally more adversarial than the per-window detection protocol that produced the TPC-H $J \geq 0.96$ figures in main-paper §H Table 12 (A14 row): all-pairs evaluates discrimination across every within/cross combination, whereas per-window detection collapses each window into a single trigger decision against ground-truth phase boundaries. Both protocols yield θ^* in the $[0.6, 0.8]$ band, supporting Theorem 3 (Economic Optimality)'s parameter-invariance prediction (operating-point invariance up to data-model normalisation).

B4 (Amplitude Invariance). To confirm that the angular kernel is truly amplitude-invariant, we scale the receipt workload by factors $k \in \{0.5, 1, 2, 5, 10\}$ and measure the resulting HSM score dimensions. The rank (S_R) and temporal (S_T) dimensions should remain constant, while the volume dimension (S_V) varies proportionally to k.

B5 (Weight and Noise Robustness). To assess HSM's robustness to simultaneous weight and noise perturbations, we construct a 2D grid varying noise level z and weight perturbation magnitude δ independently. We track both the optimal threshold θ^* (which should remain stable) and the AUC (which may degrade at large perturbations).

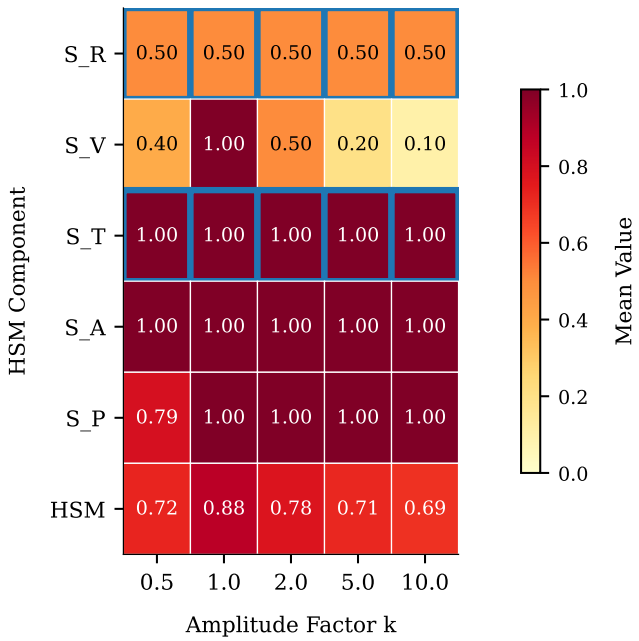

Figure 5: In-phase amplitude invariance. S_R and S_T remain constant across amplitude scaling $k \in \{0.5, 1, 2, 5, 10\}$ (10 blocks each), confirming angular kernel insensitivity to workload volume changes within a phase. S_V varies with k as expected. HSM floor ranges from 0.695 to 0.875.

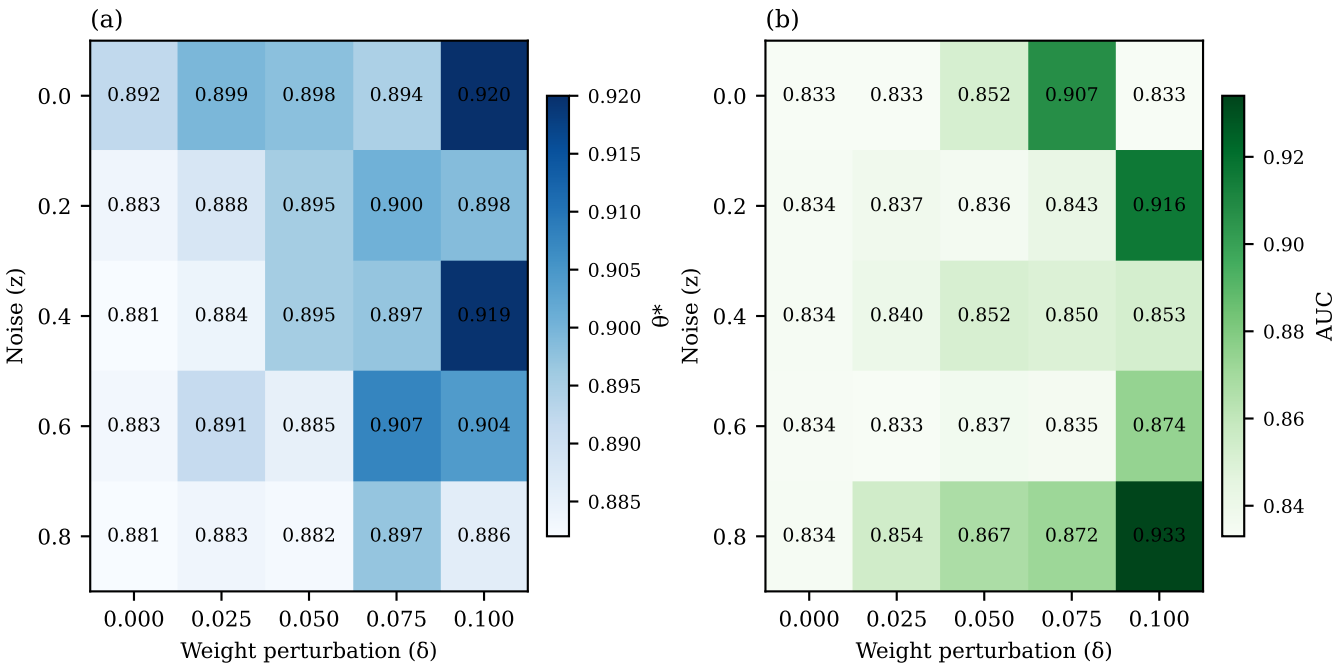

Figure 6: Robustness grid: (a) θ^* and (b) AUC across noise level z and weight perturbation δ. θ^* remains stable (0.882–0.920, CV<1%) across all perturbations. AUC degrades at $\delta \geq 0.075$.

XV. A13 Decision-Layer Cost (Full Tables and Figures)

This section reports the full per-SF wall-QPS table and the throughput-comparison figure summarised in main paper §A13, plus the cross-workload speedup-bound table (A13c) and the kernel-ablation figure for the burst-mid-Q workload.

Table 14: Decision-layer cost measurement (wall-QPS ± s.d., 10-block RCB, PostgreSQL 16). All conditions execute on identical schemas; the wall-clock gap from *Base* isolates the in-band cost of each invocation policy (decision layer + triggered advisor work). Sp._{A} = HSM-gated ratio over always-on; Sp._{P} = HSM-gated ratio over periodic ($K = 3$). Both ratios are scale-invariant (CV < 7 %) across the 15× data-volume range.

SF	Base	Always	Period.	HSM	Sp._{A}	Sp._{P}
0.2	23.19±0.56	4.81±0.17	10.03±0.81	14.36±0.88	2.99×	1.43×
1.0	4.10±0.06	0.81±0.04	1.70±0.15	2.49±0.15	3.05×	1.46×
3.0	1.06±0.01	0.25±0.01	0.48±0.04	0.67±0.05	2.71×	1.39×
HSM-gated retention (vs. Base, paired)				61.9 %	(±1.2 % across SF)	

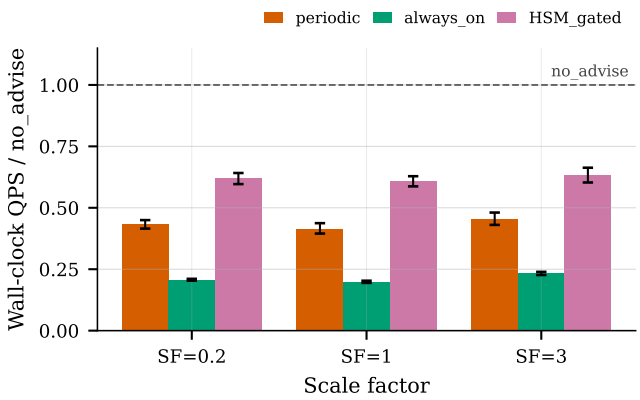

Figure 7: Decision-layer cost across invocation policies (10-block RCB, $n = 10$ per cell). All conditions execute on identical schemas; bars isolate the in-band cost of each policy (decision layer + triggered advisor work) as a within-block paired ratio versus the no-advisor baseline (95 % percentile bootstrap CI). HSM-gated retains 61.9±1.2 % of baseline at every SF (1.43× over periodic, 2.92× over always-on) while invoking the advisor only 19.2 % of the time—a scale-invariant operating point across the 15× data-volume range.

Notation Summary

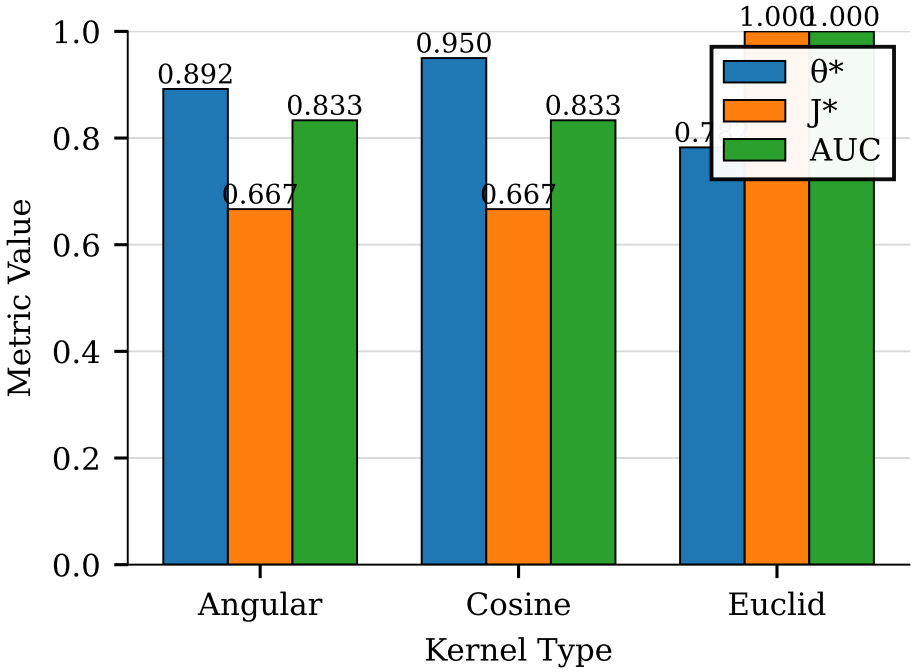

Figure 8: Kernel ablation on the burst-mid-Q workload ($n_w = 4170$, $n_c = 20$, 10 blocks). Angular and cosine kernels achieve identical $J^* = 0.667$ and AUC $= 0.833$ at $\theta^* \approx 0.9$. The Euclidean distance attains $J^* = 1.0$ (AUC $= 1.0$), an outcome consistent with overfitting to the limited cross-phase sample. Fig. 5 confirms amplitude invariance across workload scaling, and Fig. 6 validates robustness to weight perturbations.

Table 15: A13c: Cross-workload empirical validation of Theorem 5 (operational speedup with phase-mix prior). Each row is one workload at its calibrated θ^*. N = number of consecutive window-pairs; $\hat{p}^{\mathrm{T}}_{\mathrm{stable}}$ = fraction of within-phase pairs; Bnd = $1/(1-\hat{p}^{\mathrm{T}}_{\mathrm{stable}})$ (asymptotic upper bound of Theorem 5); TPR = gate sensitivity against phase labels; $\mathrm{sav_A}/\mathrm{sav_P}$ = realised invocation savings versus always-on / periodic ($K = 3$). TPC-H entries aggregate 10-block RCB across SF $\in \{0.2, 1.0, 3.0\}$.

Workload	θ	N	$\hat{p}^{\mathrm{T}}_{\mathrm{stable}}$	Bnd	TPR	$\mathrm{sav_A}$	$\mathrm{sav_P}$
TPC-H (3 SFs)	0.750	240	—	—	—	80.8 %	42.5 %
Burst	0.750	26	0.923	13.0×	0.50	80.8 %	44.4 %
SDSS (θ^*)	0.633	4954	0.731	3.71×	0.60	75.1 %	25.2 %
JOB (θ^*, small n)	0.728	20	0.850	6.67×	0.00	95.0 %	85.7 %
OLTP (pgbench)	0.750	37	0.919	12.3×	1.00	0.0 %	—

Table 16: Notation used in Supplementary Sections I–XIII (Lemmas 1–4, Theorems 1–5, extended experiments, and additional analyses).

Symbol	Definition
W_A, W_B	Comparison workload windows
$N_{\mathrm{pts}} = \lvert W_A\rvert + \lvert W_B\rvert$	Total query events across both windows
N	Database cardinality (rows in indexed table)
S_R, S_V, S_T, S_A, S_P	Five HSM dimension scores, each $\in [0,1]$
$w_R = 0.25;\ w_V = w_T = w_A = 0.20;\ w_P = 0.15$	Default dimension weights
$d_{\mathrm{HSM}} = 1 - S_{\mathrm{HSM}}$	HSM distance function
θ^*	Optimal RETAIN/REBUILD decision threshold (Theorem 3)
p_{stable}	Probability HSM decides RETAIN on a window
$\hat{p}_m$	Empirical RETAIN rate from m window observations
$\eta_{m,\delta}$	Hoeffding half-width $\sqrt{\log(2/\delta)/(2m)}$
$C_{\mathrm{db4}} = \sqrt{2}/2$	db4 ℓ^2-norm filter constant (Lemma 4)
$\varepsilon(M, N_{\mathrm{pts}}, L)$	DWT pattern-preservation error (Theorem 1)
$Q_{\min}(N)$	Break-even query volume (Theorem 3)

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